

Agent Workspace as Code v0.14.x: A Governed Causal Control Plane for Production Agentic AI

Executive summary

This white paper presents a source-grounded v1 draft of **Agent Workspace as Code** as a production architecture for agentic systems. In the public materials reviewed, Growthhub Local defines AWaC as an owned workspace artifact composed of the app surface, validated configuration, data model, fork metadata, trace, workflows, source records, and optional hosted authority. Public repo and product materials verify a local-first governed workspace stack, a workspace builder runtime, governed data objects, API Registry and resolver lanes, sandbox execution, browser authority on `sandbox-environment` rows, governed swarm workflows, and a CEO oversight cockpit. They also verify that the public branch snapshot reviewed for this paper shows `@growthhub/cli@0.14.4`, `@growthhub/create-growthhub-local@0.14.4`, and `@growthhub/api-contract@1.5.0`, with the repo explicitly warning contributors to treat package manifests—not prose—as the version source of truth. Accordingly, this paper uses the requested **v0.14.x** framing, but treats any finer-grained claims about **0.14.5** or **API Contract 1.5.1** as **internal or unspecified** unless independently verifiable in public sources. Likewise, **0.13.7** is treated as historical context supplied by the requester rather than a publicly corroborated release point; the earliest public “current shipped reality” item surfaced in the reviewed README is **0.13.9**. ¹

The central thesis of this draft is that AWaC is best understood not as a prompt wrapper or orchestration convenience layer, but as a **governed causal control plane**. Public Growthhub documentation repeatedly describes the workspace as the source of truth, the API as the mutation boundary, source records as evidence, workflows as executable rows, the helper as a proposal engine, and Workspace Lens or the CEO cockpit as readiness and oversight surfaces. This paper formalizes those patterns into a causal model in which rows are treated as causal seeds, state derivation is kept pure where possible, and mutation moves through proposal, review, apply, execution, and receipt-backed state update loops. ²

The paper also situates AWaC in the broader 2025–2026 industry and research landscape. Official announcements from Dynatrace, AWS, and Duco show a convergent market shift toward agentic operations, workspace-centric collaboration, and deterministic or evidence-grounded automation. At the research level, Causely, CausalPulse, AgentTrace, CausalReasoningBenchmark, PROV-AGENT, W3C PROV, OpenTelemetry, MCP, A2A, and the OpenAI Agents SDK all reinforce the same design direction: trusted agentic systems need structured state, provenance, explicit capability boundaries, interoperable protocols, and observable execution rather than prompt-only autonomy. ³

The appended research extension reframes AGI talk into a more defensible hypothesis: **emergent operational intelligence** may arise when governed workspaces accumulate causal density, resonance, evidence completeness, capability trust, and governance strength. That claim is framed as a testable research program rather than a declaration of achieved general intelligence. ⁴

Lineage and positioning

A corrected public reading of the lineage matters because the public repo itself warns that version numbers in prose and screenshots go stale, and instructs contributors to read package manifests as the authoritative source. In the public README snapshot reviewed for this white paper, Growthhub Local presents the “current shipped reality” as a coherent AWaC surface with public milestones for **0.13.9**, **0.14.0**, **0.14.1**, **0.14.2**, and **0.14.4**. The same README states that the then-current public package alignment is `@growthhub/cli@0.14.4`, `@growthhub/create-growthhub-local@0.14.4`, and `@growthhub/api-contract@1.5.0`. Publicly retrievable materials reviewed for this paper did **not** independently verify a public `0.14.5` or `1.5.1` artifact, so those later deltas are best treated as internal or unpublished at the time of review. The user-requested reference to **0.13.7** likewise could not be corroborated in public repo materials; the earliest public point enumerated in the repo’s current-line summary is **0.13.9**. ⁵

That corrected lineage still supports the white paper’s broader thesis. Public Growthhub materials verify a progression from governed workspace creation and API onboarding in **0.14.0**, to governed swarm workflows in **0.14.1**, to controlled browser authority for sandboxes and swarm nodes in **0.14.2**, to a CEO oversight cockpit in **0.14.4**. This is already enough to describe a real architectural arc: the workspace evolves from configurable artifact to operational control plane, with each release tightening the link between state, capability, execution, and evidence. ⁶

Industry context reinforces that interpretation. Dynatrace now positions **Dynatrace Intelligence** as an “agentic operations system” that fuses deterministic AI with agentic action, grounded by Smartscape’s real-time dependency graph and Grail’s unified lakehouse. AWS positions **Amazon Quick** as an AI assistant that can now create documents and visuals in chat and, as of June 2026, run autonomous agents with adjustable autonomy and unified multi-dataset analytics. Duco describes **agentic workspace** as a controlled collaboration layer where humans and intelligent agents work inside a single auditable operational environment. These products are not identical to AWaC, but together they show that enterprise AI is moving toward governed, context-rich operating layers rather than isolated chat surfaces. ⁷

Public Growthhub architecture documents make the local-versus-hosted split explicit. The system is described as two cooperating layers: a **local execution layer** for environment creation, templates, workflows, fork state, and local harnesses, and an **optional hosted authority layer** for identity, connection authority, capability activation gates, and hosted workflow or integration bridges. The product site describes the same local-first posture, noting that Growthhub Local is in developer preview, deploys a Next.js workspace app surface, and keeps baseline value usable without a mandatory hosted dependency. That is strong public evidence for a control-plane trajectory, even though public sources reviewed here do not yet document a full fleet-management console across many workspaces in detail. ⁸

Architecture and causal control plane

The architectural primitive in AWaC is the **workspace artifact**. Public documentation defines AWaC as the idea that the workspace itself is the owned product object, not just the UI or the CLI. The artifact includes `growthhub.config.json`, the `apps/workspace` app surface, `.growthhub-fork/` identity and policy state, data model objects, source records, helper receipts, workflow and sandbox rows, resolver files, skills, helpers, templates, docs, and trace. Growthhub’s operating-framework guide collapses the mental model

into a short rule: **state** → **eligibility** → **guidance** → **action** → **evidence** → **next state**. That is precisely the sort of lifecycle one would expect from a causal control plane rather than a chat-centric orchestrator. ⁹

Public contracts also show why **rows can be treated analytically as causal seeds**.

`dataModel.objects[]` is the canonical local surface for business objects; rows are not accidental UI state, but the carriers of durable object identity and governed mutation. The same surface holds special rows such as `sandbox-environment`, which define where and how workloads run, including locality, adapter, optional scheduler link, and browser authority. Workflows are not hidden task runners; they are sandbox rows with orchestration state. Source records are sidecar-backed evidence that hydrate live-backed objects at runtime without smuggling mutable external state into the core config. In other words, a row in AWaC is not just a record. It is a stable locus of authority, evidence, and future affordance. ¹⁰

Figure 1. Causal control loop

```
flowchart LR
    S[Workspace state W_t] --> D[D[Eligibility and derived view F(W_t)]]
    D --> P[P[Proposal]]
    P --> R[R[Review]]
    R --> A[A[Apply]]
    A --> X[X[Execution]]
    X --> E[E[Receipt and source record]]
    E --> N[N[Updated state W_{t+1}]]
    N --> D
```

Publishing note: export Figure 1 to SVG and PNG from this Mermaid source during layout.

The current public contracts do not themselves name a **dual-graph stack**, but they strongly support modeling the system that way. This white paper uses the term to distinguish a **context graph** from a **decision graph**. The context graph contains business rows, sandbox rows, resolver and API rows, source records, receipts, and readiness state. The decision graph contains the publish-owned `orchestrationGraph` and `orchestrationConfig` used for workflow and swarm execution. That separation is not cosmetic. The topology contract explicitly states that live workflow state is publish-owned, draft-to-live transitions occur through `POST /api/workspace/workflow/publish`, and direct live-state mutation through PATCH is rejected. Meanwhile, the runtime contract describes pure row-to-values computation for chart projections and read-only source-record hydration. This suggests a principled split between **what the workspace knows** and **how the workspace decides or executes**. ¹¹

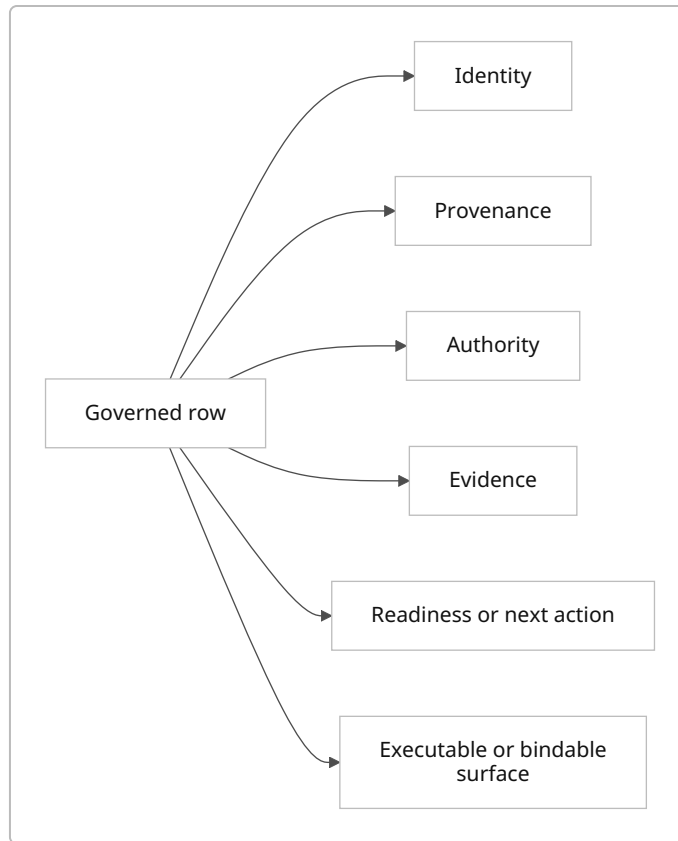
Figure 2. Dual-graph stack



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The case for **pure causality-driven functions** is public and concrete, not merely aspirational. The builder runtime documents `computeChartValuesFromRows` as a **pure module**: no browser fetch, no provider logic, no secrets, and no schema mutation. It projects rows into finite chart values and exposes a companion debug helper that returns diagnostics without becoming a parallel runtime. That offers a general architectural pattern: use pure derivation functions for state interpretation, then gate all side effects behind governed mutation and execution routes. In a stricter formalization, AWaC can be described by a workspace state `W_t`, a pure derivation function `Derived_t = F(W_t)`, and a mutation loop that only changes the artifact through explicit governed transitions. Public docs do not claim full system-wide referential purity, but they clearly support purity as a design norm for derivation and read models.

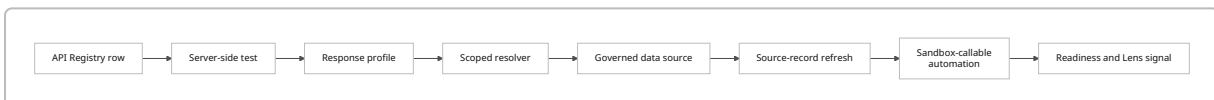
Figure 3. Row as causal seed



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The public **API Registry cockpit** is the clearest evidence for capability causality. Growthub's operating framework describes an API lane that starts by registering an API row, then configuring auth references, testing server-side, profiling the response, optionally creating a resolver, creating a governed data source, refreshing source records, and confirming that sandbox automation can call the API. The "new reality" value map describes the same path as **API Registry → server-side test → response profile → optional helper resolver → governed Data Source → source-record refresh → sandbox automation → workflow persistence path → Workspace Lens readiness**. That is not a flat tool catalog. It is a capability-evidence chain. Because public sources reviewed here still point to `@growthub/api-contract@1.5.0`, any claim specific to an external **API Contract 1.5.1** should be treated as internal or unspecified, but the verified public resolver lane is already sufficient to reason about a governed capability registry. ¹³

Figure 4. API capability causality



Publishing note: export Figure 4 to SVG and PNG from this Mermaid source during layout.

Table 1 maps the major AWaC primitives to mature research and standards categories. The purpose is analytical: it shows that AWaC’s architecture is not an island, but a synthesis of ideas that already exist in provenance modeling, observability, protocol design, agent governance, and causal diagnostics. ¹⁴

Table 1. Mapping AWaC primitives to external research categories

AWaC primitive	AWaC role	External research category	Representative source
Workspace artifact	Portable, governed product object	Local-first system packaging and durable artifacts	Growthhub architecture and README; local-first software tradition ¹⁵
Data model rows	Durable business and execution state	Structured world-state / causal substrate	Growthhub topology and operating framework; Causely ¹⁶
Source records and Agent Outcomes	Evidence and continuity	Provenance and execution traceability	W3C PROV, PROV-AGENT, OpenAI tracing ¹⁷
API Registry and resolvers	Capability onboarding and trust	Tool certification / protocol-mediated capability access	Growthhub operating framework, MCP, A2A ¹⁸
sandbox-environment rows	Execution boundary and authority	Sandboxed runtime and bounded actuation	Growthhub topology and browser-access docs; OpenAI guardrails/tracing as adjacent control concepts ¹⁹
agent-swarm-v1 orchestration graph	Governed multi-agent execution	Multi-agent orchestration and causal debugging	Growthhub swarm docs; AgentTrace; CausalPulse ²⁰
Proposal, review, apply	Governed mutation lane	Human-in-the-loop control and safe actuation	Growthhub value map; OpenAI guardrails ²¹
Workspace Lens and CEO cockpit	Readiness and executive oversight	Executive control, observability, and metacognitive surfaces	Growthhub CEO and value-map docs; Dynatrace Intelligent operations surfaces ²²

Governance, operations, and deployment

Governance in AWaC is not an add-on. Public docs make it the default mutation grammar. The workspace helper is explicitly **proposal-first and apply-second**; helper queries do not write config, accepted proposals create receipts, and explicit apply is required before mutation. The “new reality” value map states the same principle in plain language: the helper remains useful in production-like workflows because it is propose-first, apply-second, credentials are stripped from prompt context, and accepted changes create receipts. This is a classic control-plane pattern: planning can be probabilistic, but mutation must pass through bounded transitions. ²³

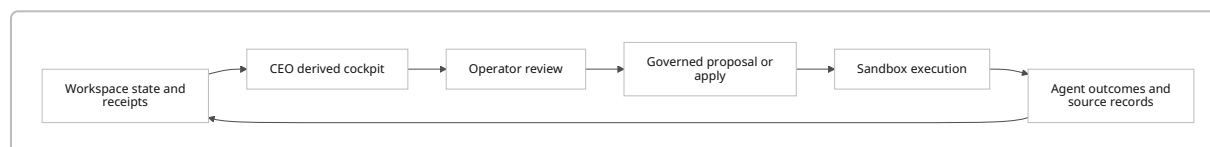
The **mutation firewall** is publicly validated through contract language even if the exact internal phrase “mutation firewall” is not surfaced in every public document. The topology contract states that `PATCH /api/workspace` is limited to specific surfaces, rejects full-config bodies, sidecar writes, oversized rows or node-configs, history blobs, credential-shaped fields, and direct live-workflow mutations, and supports a preflight dry run through `POST /api/workspace/patch/preflight`. It also states that the only route from draft to live workflow state is the server-authoritative publish endpoint, verified against sandbox-run lineage rather than user-writable attestation fields. In practice, that is a mutation firewall: it prevents arbitrary writes, separates derivation from actuation, and forces side effects into reviewable lanes. ²⁴

Sandbox and browser authority are likewise row-governed. Public browser-access docs state that browser capability is not a separate runtime or hidden model; it is a capability on a governed `sandbox-environment` row. The row carries `browserAccess`, `networkAllow`, adapter choice, and scheduler linkage. Browser authority then travels through the existing `sandbox-run` path, with node-level network permissions gating inheritance in workflow or swarm graphs. Local process adapters receive an environment contract, local intelligence uses the local browser bridge, and serverless handlers receive the same capability signal in the `growthhub-sandbox-run-v1` envelope. This is a strong design: it ties authority to durable workspace state instead of to prompt instructions or temporary session flags. ²⁵

Governed swarm execution follows the same principle. In public docs, a swarm is not a loose chat metaphor but a runnable workflow row whose graph uses `agent-swarm-v1`, whose execution is routed through `POST /api/workspace/sandbox-run`, whose telemetry is truthful rather than estimated, and whose run history is persisted as source-record evidence. The swarm cockpit’s invariants explicitly forbid fake token counts, fallback redirects, or separate persistence hacks. That aligns with current research on agent observability: OpenAI’s Agents SDK captures handoffs, generations, function-tool calls, and guardrails in traces; OpenTelemetry frames traces as structured causal paths; and AgentTrace argues that multi-agent failures require causal graph reconstruction for root-cause localization. AWaC’s public swarm model fits squarely into that broader move toward structured, inspectable multi-agent execution. ²⁶

The **Workspace CEO cockpit** is the architecture’s most explicit executive-control surface. Public docs describe `/ceo` as the operator cockpit inside the helper sidecar, not as a separate runtime. It closes the loop between first-use setup, governed swarm proposal and apply, sandbox-run proof, Agent Outcome receipts, runtime history, reusable Agent Teams, and the linked workflow canvas. The source-of-truth table for the feature is especially revealing: bootstrap state is derived from workspace config plus receipts; operational history is derived from swarm workflows plus receipts; execution still routes through `POST /api/workspace/sandbox-run`; and the workflow canvas remains the executable graph authority. In cognitive terms, that is an executive function surface: it compresses complex workspace state into safe next actions, without creating a second hidden control path. ²⁷

Figure 5. CEO cockpit loop



Publishing note: export Figure 5 to SVG and PNG from this Mermaid source during layout.

AWaC’s **templates-as-executable-OS** framing is best understood as a design interpretation grounded in public docs. Growthhub’s architecture describes worker kits as **portable execution environments**, not just prompts. The README describes templates as reusable workspace layouts and kits as self-contained environments with assumptions, examples, helpers, and output standards. The operating framework then shows that the workspace can be entered through starter, repo-import, skill-import, or kit-download paths, all normalized into the same governed shape. That makes templates and kits more than scaffolds; they are the boot surfaces of repeatable operational environments. In that limited and defensible sense, templates behave like an executable operating substrate for agentic workspaces. ²⁸

Drift detection is another essential production property. Growthhub’s README and architecture docs describe self-healing forks with drift detection, dry-run heal plans, protected paths, append-only trace logging, and additive updates designed to preserve customization without losing the upgrade path. That directly addresses a common failure mode in agentic environments: operational configuration copied by hand until it slowly diverges. Public docs do not yet expose a formal academic drift metric, but the design is recognizably analogous to infrastructure-as-code control layers in which declarative state, preview-before-apply, protected paths, and append-only traces keep environments reproducible over time. ²⁹

Deployment is explicitly **local-first, hosted-authority-optional**. Public sources describe the workspace as a deployable Next.js app with Vercel-ready checks, while the topology contract notes that filesystem writes are opt-in outside dev runtimes and default to read-only on serverless deploys. Hosted authority is additive: useful for account-backed integrations, stronger execution, and managed team workflows, but not the front-line operating center of the workspace. For enterprise deployment, that implies a practical pattern: keep the control plane artifact local and portable, use serverless and hosted authority where stronger trust or durability is needed, and avoid shifting the source of truth away from the workspace artifact itself. Detailed public documentation for a higher-order fleet console across many workspaces remained unspecified in the reviewed sources, so this paper treats “fleet/control-plane” as an architectural direction, not a fully public product claim. ³⁰

The resulting evaluation framework should measure not just model quality but the quality of the governed causal plane around the model. Table 2 proposes a practical metric set grounded in the measurement families used by Causely, CausalPulse, AgentTrace, OpenAI tracing, and provenance-based workflow research. ³¹

Table 2. Evaluation metrics for AWaC

Metric	Measurement method	Why it matters	Suggested v1 target
Workflow success rate	Successful end-to-end runs ÷ attempted runs	Baseline production reliability	≥ 95% on bounded workflows
Root-cause or task-correctness rate	Human-verified correctness on incident or business tasks	Captures actual task value, not just completion	≥ 90% in controlled evals
Time to diagnosis or completion	Median time from run start to correct result	Directly comparable to Causely and CausalPulse-style benchmarks	30–50% improvement over unguided baseline

Metric	Measurement method	Why it matters	Suggested v1 target
Token footprint	Median tokens per successful run	Measures semantic-interpretation tax	25–50% reduction versus unguided baseline
Tool-call footprint	Median tool calls per successful run	Measures orchestration efficiency	20–50% reduction versus unguided baseline
Receipt coverage	State-changing actions with receipts ÷ all state-changing actions	Core trust and replayability indicator	≥ 99%
Resolver trust coverage	Trusted resolvers ÷ resolver references used in workflows	Detects capability hallucination risk	≥ 95%
Drift exposure window	Median time between upstream drift and detection	Measures configuration hygiene	< 24 hours in CI-backed environments
Publish safety rate	Drafts published only after passing required tests	Captures control-plane integrity	≥ 99%
Rollback or replay success	Successful replay or rollback attempts ÷ requested attempts	Measures recovery quality	≥ 95%
Operator review burden	Median reviews per governed mutation; NASA-TLX or similar subjective load	Ensures governance is usable, not suffocating	Declining per mature workspace
Evidence freshness	Median age of linked source records powering decisions	Captures stale-state risk	Domain-specific; ideally < SLA window

Emergent operational intelligence

The research extension proposed here adopts a deliberately careful thesis: **AWaC may constitute a practical substrate for emergent operational intelligence**, not a proof of AGI. That claim becomes plausible when one reads Growthub’s public materials alongside 2025–2026 causal-AI and agentic-systems research. Growthub explicitly centers governed state, source-record evidence, proposal-bound mutation, workflow graphs, readiness surfaces, and reuse of the same artifact by humans and agents. Causely shows that giving agents a live causal model of environment topology can cut diagnosis time, token use, and tool calls while improving accuracy. CausalPulse shows that a multi-agent causal-diagnostics copilot can achieve very high reliability in industrial deployment. AgentTrace shows that post-hoc causal graph reconstruction is practical for deployed multi-agent systems. PROV-AGENT, W3C PROV, OpenTelemetry, and the OpenAI Agents SDK all converge on the need for formalized provenance and runtime observability. ³²

From that evidence, seven patterns emerge.

First, **causal resonance** describes the degree to which one workspace element explains another. Growthhub's own public state model already implies this: a `browserAccess` bit on a sandbox row explains node-level execution authority; a tested API row explains whether source refresh is meaningful; a receipt-backed workflow status explains whether a publish step is safe. Causely's benchmark can be read as an external validation of the same principle: when topology and causal relationships are structured rather than inferred ad hoc from raw telemetry, investigation becomes faster, leaner, and more accurate. ³³

Second, **capability self-certification** arises when a declared capability progresses through an evidence chain rather than existing as a bare tool description. Growthhub's public API cockpit already implements the bones of this pattern: register, test, profile, optionally create resolver, create data source, refresh source records, then confirm sandbox automation. MCP and A2A matter here because they make capability transport and inter-agent interoperability more standardized, while OpenAI guardrails show how execution boundaries can be enforced around tool or workflow invocation. ³⁴

Third, **governed swarm self-modeling** is possible because public AWaC swarm flows preserve both the orchestration graph and the evidence trail. A swarm can be analyzed in terms of its graph, its source-record run history, its truthful telemetry, and its replayable row identity. AgentTrace's findings strengthen this framing by showing that graph-centric diagnosis is better suited than naive trace inspection for complex multi-agent failures. ³⁵

Fourth, **executive meta-governance** is embodied in Workspace Lens and the CEO cockpit. Public docs describe these not as alternative runtimes but as read models over the same artifact, configured to answer what is ready, what is blocked, what has evidence, and what should happen next. In other words, governance becomes governable. That is a significant systems property because it lets operators reason about the control plane itself instead of merely about the domain task. ³⁶

Fifth, **self-evidencing execution** follows from receipts and source records. W3C PROV defines provenance as information about entities, activities, and people involved in producing a thing, enabling judgments about reliability and trustworthiness. PROV-AGENT extends that logic to agent workflows. Growthhub's public docs fit naturally into that lineage: source records are the evidence layer, helper apply creates receipts, and runtime outcomes feed both workflow and CEO surfaces. AWaC does not prove all truth internally, but it is architected to persist enough linked evidence that later humans and agents can verify what happened. ³⁷

Sixth, **homeostatic mutation control** appears when invalid transitions are rejected before they destabilize the workspace. Public AWaC patch policy, draft/test/publish workflow rules, and browser-authority boundaries all point in this direction. OpenAI guardrails provide an external analog at the workflow level: input, output, and tool guardrails are explicit checks around reasoning and tool use. AWaC's contribution is to push the same idea into the workspace artifact and its mutation surfaces. ³⁸

Seventh, **emergent operational intelligence** is the paper's hypothesis that performance can improve as workspace structure improves, even without changing the underlying model. CausalReasoningBenchmark is instructive here: strong models may identify broad causal strategy, yet fail on detailed specification. That suggests a design opportunity for a better substrate. If more of the identification burden is carried by governed workspace state, trusted capabilities, and evidence-backed context, then the model's job shifts from hallucination-prone reconstruction to bounded planning over a better world model. ³⁹

A minimal formalization of the research hypothesis is:

$$I_{workspace} \propto D_{causal} \times R_{resonance} \times E_{evidence} \times T_{capability} \times G_{governance}$$

where `D_causal` is the density of meaningful links among rows, graphs, receipts, and sources; `R_resonance` is the informational explanatory power of one state element over connected state; `E_evidence` is receipt and provenance completeness; `T_capability` is the ratio of actually trusted to merely declared capabilities; and `G_governance` captures the strength of the review, publish, boundary, and rollback controls. This is a **speculative scaling law**, not a validated one. Its value lies in making AWaC a measurable research program rather than a slogan. Research on Causely, CausalPulse, and AgentTrace suggests that at least three of these variables—causal structure, evidence richness, and run observability—already move real performance. ⁴⁰

Experimental agenda

The experimental agenda below translates the research extension into a concrete program. The design draws on measurement families already used in Causely, CausalPulse, AgentTrace, PROV-AGENT, OpenTelemetry, and the OpenAI Agents SDK, but the success criteria here are proposed for an AWaC-specific program rather than copied from any one benchmark. ³¹

Table 3. Proposed AWaC experimental agenda

Experiment	Objective	Setup	Variables	Metrics	Hypothesis	Success criteria
Lineage reproducibility	Verify that identical workspace artifacts reproduce the same governed surfaces across environments	Same workspace on two clean machines and one serverless deploy	Runtime, persistence mode, branch	QA pass rate, diff count, deploy-check pass rate	Artifactized AWaC reduces setup variance	Zero unexplained config diffs; >95% identical readiness outputs
Pure derivation determinism	Test whether derived readiness and chart projections remain deterministic	Fixed row sets and receipts replayed repeatedly	Input order, missing data, stale sidecar data	Function output equality, warning consistency	Pure derivations remain stable under replay	100% identical outputs for identical input state

Experiment	Objective	Setup	Variables	Metrics	Hypothesis	Success criteria
Proposal-review-apply safety	Measure governance overhead and mutation correctness	Operators use helper on standard setup tasks	Review requirement, task complexity	Mutation error rate, review time, operator load	Reviewable mutations reduce bad writes without killing usability	$\geq 50\%$ fewer invalid writes with $< 20\%$ time overhead
Resolver self-certification	Compare declared-tool vs evidence-backed resolver planning	Same tasks under plain tool lists and API-cockpit-certified resolvers	Trust surface, auth readiness, response profiling	Cap-hallucination rate, successful API calls, task completion	Certified capabilities cut hallucinated tool use	$\geq 40\%$ reduction in bad tool assumptions
Source-record freshness	Quantify impact of runtime sidecar evidence on dashboard and workflow correctness	Live-backed objects refreshed at varying cadences	Refresh interval, source volatility	Freshness age, dashboard mismatch rate, wrong-action rate	Earlier evidence reduces false operational conclusions	Mismatch rate $< 2\%$ at target SLA
Browser authority containment	Validate that browser actions only occur when governed grants allow them	Mix of sandbox rows with/without <code>browserAccess</code> and node network permission	Adapter type, node policy	Unauthorized-browser success, logged denials, task pass rate	Row-governed authority contains blast radius	0 unauthorized browser executions
Governed swarm repeatability	Compare freeform prompt swarms against governed <code>agent-swarm-v1</code> rows	Same multi-agent tasks across both modes	Graph availability, truthful telemetry, row ownership	Success rate, recovery time, audit completeness	Governed swarms are more repeatable and auditable	$\geq 20\%$ better repeatability; full audit trail on governed runs

Experiment	Objective	Setup	Variables	Metrics	Hypothesis	Success criteria
CEO cockpit operator value	Evaluate whether <code>/ceo</code> improves human-agent coordination	Operators perform setup, incident review, and handoff tasks with and without CEO cockpit	Surface used, user expertise	Time to next safe action, confidence, NASA-TLX	Derived cockpit reduces cognitive load	$\geq 25\%$ faster safe-next-action selection
Receipt coverage and continuation	Test whether future agents continue more accurately with richer receipt coverage	Execute workflows, then ask fresh agents to continue or repair	Receipt coverage 0–100%	Duplication rate, false assumptions, repair success	Continuation quality rises sharply with full evidence	$>90\%$ correct continuation above 90% coverage
Causal density scaling	Measure relation between richer workspace linkage and task performance	Same tasks in low-, medium-, high-linkage workspace variants	Link density among rows, sources, workflows	Task success, tool calls, token use, reasoning steps	More causal structure improves bounded planning	Monotonic gains until complexity saturation point
Resonance scaling	Test whether explanatory state reduces semantic tax	Agents operate on raw logs versus derived readiness/trust projections	Projection richness, state compression	Time to diagnosis, token footprint, tool-call count	High resonance lowers interpretation cost	$\geq 25\%$ reduction in tokens and tools
Fleet generalization	Evaluate how well mature workspaces generalize to new but adjacent tasks across many artifacts	10–20 workspaces built from same starter family	Task novelty, workspace maturity, shared resolver set	Transfer success, edit distance, new-code requirement	Structured workspaces generalize by recombination	$>70\%$ success on unseen adjacent tasks without new code

References and appendix

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Appendix

The formal model used in this white paper is intentionally narrower than the product’s full implementation surface. A minimal analytical state can be written as:

```
W_t = {
  C_t // workspace config and topology
  D_t // data-model objects and rows
  S_t // sandbox-environment rows
  G_t // orchestration graphs and workflow state
  R_t // API Registry and resolver state
  O_t // source records, helper receipts, agent outcomes
  P_t // proposal/review/apply and publish boundaries
  H_t // cockpit and readiness projections
}
Derived_t = F(W_t)
W_t -> Proposal_t -> Review_t -> Apply_t -> Execute_t -> Receipt_t -> W_t+1
```

The paper’s proposed research metrics can be formalized as:

```
CausalDensity(W) =
  meaningful_links(W) / possible_links(W)

Resonance(W) =
  1 - H(W | E) / H(W)

EvidenceCompleteness(W) =
  receipt_backed_state_changes / all_state_changes

CapabilityTrust(W) =
  trusted_resolvers_used / resolvers_referenced
```

```
GovernanceStrength(W) =  
    bounded_transitions_enforced / bounded_transitions_required
```

A practical instrumentation stack for AWaC should combine OpenTelemetry-style traces for execution paths, PROV-style entity/activity/agent provenance for artifact mutations, and an AWaC-native receipt schema that links workspace objects to runtime outcomes. At minimum, each receipt should capture `workspaceId`, `objectId`, `rowName`, `actionType`, `proposalId`, `applyId`, `executionId`, `inputRefs`, `outputRefs`, `resolverRefs`, `sandboxRef`, `operatorRef`, `status`, `startedAt`, `completedAt`, `draftHash`, `publishedHash`, and `nextActionHint`. That schema would let traces answer “what happened,” provenance answer “what produced what,” and the control plane answer “what is safe to do next.” ⁶⁷

A proposed **resolver trust score** can be implemented as a weighted composite rather than a binary label:

```
ResolverTrust =  
    0.25 * auth_readiness  
+ 0.20 * test_success  
+ 0.15 * schema_profile_quality  
+ 0.15 * runtime_registration  
+ 0.15 * recent_success_rate  
+ 0.10 * receipt_coverage
```

Finally, the white paper’s central methodological recommendation is simple: evaluate AWaC as a **system**. Measure the model, but also measure the artifact, the mutation boundaries, the provenance layer, the capability registry, the swarm graph, and the executive cockpit. The strongest evidence in the reviewed literature suggests that production agent quality is driven not by models alone, but by the quality of the causal environment in which those models operate. ³¹

¹ ⁶ ²⁹ [growthhub-local/README.md](https://github.com/Growthhub-ai/growthhub-local/blob/main/README.md) at main · Growthhub-ai/growthhub-local · GitHub
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² ⁹ ¹³ ¹⁸ ²³ ³² ³⁴ ⁵¹ [growthhub-local/docs/AGENTIC_WORKSPACE_AS_CODE_OPERATING_FRAMEWORK.md](https://github.com/Growthhub-ai/growthhub-local/blob/main/docs/AGENTIC_WORKSPACE_AS_CODE_OPERATING_FRAMEWORK.md) at main · Growthhub-ai/growthhub-local · GitHub
https://github.com/Growthhub-ai/growthhub-local/blob/main/docs/AGENTIC_WORKSPACE_AS_CODE_OPERATING_FRAMEWORK.md

³ ⁷ ⁵³ [Dynatrace Introduces Domain Specific Agents, to Extend SRE, Development, and Security Teams with Trusted, Autonomous Action](https://www.dynatrace.com/news/press-release/dynatrace-introduces-domain-specific-agents/?utm_source=chatgpt.com)
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